FLIGHTFIX

Aircraft Maintenance Efficiency — MRO Management System

SOFTWARE TESTING REPORT

Functional, Automated and Data-Driven Testing

Item	Details
Project	FlightFix: Aircraft Maintenance Efficiency — MRO Management System
Testing Framework	Django Automated Testing Framework / unittest
Application	Django Web Application
Automated Test Suite	26 test cases
Final Result	26 Passed, 0 Failed, 0 Errors
Test Execution Time	3.830 seconds
Test Database	Temporary Django test database
Prepared For	Academic Project Evaluation
Student	AYUSH SAHU (2400321540061)
Project Guide	Mr. Abhi Gurjar
Academic Year	2026–2027

1. Introduction

This Software Testing Report documents verification and validation activities performed for the FlightFix Django web application. FlightFix covers aircraft management, maintenance work orders, spare-parts inventory, technician management, authentication and dashboard access.

Testing was performed using manual functional verification and Django automated testing. Automated tests use an isolated temporary database and therefore do not modify the normal development database.

2. Testing Objectives

• Verify that major FlightFix modules behave according to their implemented requirements.

• Validate Create, Read, Update and Delete (CRUD) operations for core entities.

• Verify authentication and protected-page access.

• Validate aircraft search and status filtering.

• Verify maintenance status and aircraft relationships.

• Validate inventory stock-status logic for In Stock, Low Stock and Out of Stock conditions.

• Verify technician records and status counts.

• Confirm that important application pages return the expected HTTP responses.

• Provide repeatable automated evidence that the tested functionality continues to work.

3. Testing Scope

Module	Test Coverage
Authentication	Login, invalid login, logout, protected dashboard
Aircraft Management	Count, status counts, search, filter, create, update, delete
Maintenance	Count, pending count, aircraft relationship, status update
Inventory	Count, stock-status logic, quantity update, zero-stock condition
Technicians	Count, status counts, create
Page Access	Dashboard and major module page access after authentication
Test Data	Controlled sample aircraft, maintenance, inventory and technician records

4. Testing Strategy

Testing Type	Approach	Status
Manual Functional Testing	User-facing workflows and responsive UI were checked manually during development.	Completed
Automated Functional Testing	Django TestCase-based tests verify repeatable application behavior.	Completed
Data-Driven Testing	Controlled test records are generated in the temporary test database.	Completed
Regression Support	The automated suite can be rerun after future code changes.	Ready

5. Test Environment

Component	Configuration
Operating System	macOS
Runtime	Python 3.13.7 virtual environment
Framework	Django
Application	FlightFix
Test Runner	Django manage.py test / unittest
Database During Tests	Temporary isolated Django test database

Component	Configuration
Development IDE	Visual Studio Code

6. Test Dataset

Entity	Records	Coverage
Aircraft	3	Operational; Maintenance; Inspection Due
Maintenance	3	Scheduled; In Progress; Overdue
Inventory	3	In Stock; Low Stock; Out of Stock
Technicians	3	Available; On Task; On Leave
Test User	1	Valid authenticated user

7. Automated Test Cases

ID	Test Case	Input / Condition	Expected Result	Result
TC-01	Valid Login	Valid username and password	User is authenticated and redirected to /dashboard/	Passed
TC-02	Invalid Login	Incorrect password	Login remains unsuccessful and response is handled correctly	Passed
TC-03	Logout	Authenticated user requests logout	User is logged out and redirected to /login/	Passed
TC-04	Aircraft Count	3 aircraft test records	Aircraft count equals 3	Passed
TC-05	Aircraft Status Counts	Operational/Maintenance/Inspection Due records	Each status count is correct	Passed
TC-06	Aircraft Search	Search for FFX-T002	Matching aircraft appears and unrelated record is excluded	Passed
TC-07	Aircraft Status Filter	Filter status = Maintenance	Maintenance aircraft is returned	Passed
TC-08	Aircraft Create	Create FFX-T004	New aircraft record is stored	Passed
TC-09	Aircraft Update	Change FFX-T001 status	Updated status is persisted	Passed
TC-10	Aircraft Delete	Delete FFX-T003	Record no longer exists	Passed
TC-11	Maintenance Count	3 maintenance records	Maintenance count equals 3	Passed
TC-12	Pending Maintenance Count	Scheduled + In Progress + Overdue	Pending count equals 3	Passed
TC-13	Maintenance Relationship	Scheduled task linked to FFX-T001	Aircraft relationship is correct	Passed
TC-14	Maintenance Status Update	Scheduled → Completed	Updated status is persisted	Passed
TC-15	Inventory Count	3 inventory records	Inventory count equals 3	Passed
TC-16	Inventory Status Logic	25/10, 5/10 and 0/5 quantities	In Stock, Low Stock and Out of Stock are returned correctly	Passed
TC-17	Inventory Update	Change quantity 25 → 5	Quantity updates and status becomes Low Stock	Passed
TC-18	Zero Inventory Quantity	Quantity = 0	Status is Out of Stock	Passed
TC-19	Technician Count	3 technician records	Technician count equals 3	Passed
TC-20	Technician Status Counts	Available/On Task/On Leave	Each status count is correct	Passed
TC-21	Technician Create	Create FFX-EMP004	New technician record is stored	Passed
TC-22	Dashboard Access Control	Unauthenticated request to /dashboard/	Request is redirected to login	Passed
TC-23	Dashboard After Login	Authenticated request to /dashboard/	Dashboard returns HTTP 200	Passed
TC-24	Aircraft Page After Login	Authenticated request to /aircraft/	Aircraft page returns HTTP 200	Passed
TC-25	Maintenance Page After Login	Authenticated request to /maintenance/	Maintenance page returns HTTP 200	Passed
TC-26	Inventory/Technicians Page Access	Authenticated requests to module pages	Pages return HTTP 200	Passed

8. Automated Test Execution Summary

Metric	Result
Total test cases executed	26
Passed	26
Failed	0
Errors	0
Execution time	3.830 seconds
Overall result	PASS

26 / 26 automated tests passed successfully

The final automated execution completed successfully with no failures and no errors. Django created and destroyed the temporary test database as part of the test lifecycle.

9. Defects Identified and Resolved During Automated Testing

Defect ID	Observed Issue	Resolution	Status
D-01	Technician experience test data used text such as '5 years' while the model field expects a number.	Changed automated test data to numeric values (5, 4, 6 and 3).	Resolved
D-02	Aircraft search/filter tests initially accessed a login-protected page without authentication, producing HTTP 302 instead of HTTP 200.	Added test-user login before the protected aircraft-page requests.	Resolved

10. Manual Testing Summary

Area	Manual Verification	Status
Authentication	Login/logout workflow	Passed
Dashboard	Statistics, navigation and dashboard content	Passed
Aircraft	CRUD, search and status filter	Passed
Maintenance	Work-order CRUD and status display	Passed
Inventory	CRUD, status display and responsive table behavior	Passed
Technicians	CRUD, status display and responsive table behavior	Passed
Homepage	Hero/video, navigation and responsive layout	Passed
Responsive UI	Mobile navigation and mobile table layouts	Passed

11. Testing Evidence

The final terminal execution provides direct evidence of the automated test result. The recorded output was:

```
Ran 26 tests in 3.830s OK
```

The terminal screenshot should be retained with the project submission as supporting evidence. The automated test file is maintained at core/tests.py.

12. Conclusion

The FlightFix application completed functional verification through manual and automated testing. The final automated execution covered 26 test cases and all 26 passed successfully, with zero failures and zero errors. The suite verified authentication, protected-page access, aircraft operations, maintenance operations, inventory stock logic, technician records, relationships and core page responses.

The automated suite can be reused as a regression test suite whenever the FlightFix codebase is modified. Future releases may additionally include performance/load, security, cross-browser and API testing if those capabilities are introduced.

13. Testing Sign-Off

Field	Details
Prepared By	AYUSH SAHU (2400321540061)
Project Guide	Mr. Abhi Gurjar
Project	FlightFix: Aircraft Maintenance Efficiency — MRO Management System
Testing Status	Completed
Final Automated Result	26/26 Passed